

Minsu Kim, Ph.D.

Postdoctoral Candidate

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EDUCATION

Mar. 2019 – Feb. 2025	Konkuk University Integrated M.S. and Ph.D. in Bioscience & Biotechnology Advisor: Dr. Yoon Kyung Choi	Seoul, South Korea
Mar. 2013 – Feb. 2019	Konkuk University B.S. in Biomedical Science & Engineering	Seoul, South Korea

RESEARCH PROFILE

Neuroscientist with expertise in molecular neurobiology and computational transcriptomics.

My research integrates wet-lab neuroscience with large-scale transcriptomic analyses to investigate molecular mechanisms underlying spaceflight-induced brain adaptation and neurodegeneration. I have independently designed and led research projects using NASA GeneLab datasets, developed computational analysis workflows, and applied transcriptomic meta-analysis to identify biologically conserved molecular signatures across independent studies.

RESEARCH INTEREST

- Space Biology
- Computational Transcriptomics
- Molecular Mechanisms of Spaceflight Adaptation
- Neurodegeneration and Brain Aging
- Mitochondrial Biology and Cellular Stress
- Multi-omics Data Integration

RESEARCH EXPERTISE

EXPERIMENTAL BIOLOGY

- Mouse models of traumatic brain injury (TBI)
- Drug administration and experimental intervention
- Mammalian cell culture
 - Macrophages
 - Astrocytes
 - Neural Stem Cells
 - Endothelial Cells
 - Pericytes
- Immunofluorescence
- Western blotting
- Cryosection preparation
- Fluorescence microscopy
- Quantitative image analysis (ImageJ)

COMPUTATIONAL BIOLOGY

Transcriptomics

- Bulk RNA-seq analysis
- Differential expression analysis (DESeq2)
- Gene Set Enrichment Analysis (GSEA)
- Functional annotation & pathway analysis
- Cross-species transcriptomic integration
- Transcriptomic meta-analysis

Programming & Software Development

- OmicsForge (R): Modular platform for reproducible transcriptomics analysis and visualization.
- Salmon RNA-seq preprocessing pipeline (Python)

RESEARCH EXPERIENCES

1. INDEPENDENT RESEARCHER

(Mar. 2025 – Present)

Space Biology • Computational Transcriptomics

Designed and led independent computational research integrating publicly available transcriptomic datasets, cross-species analyses, and reproducible software development to investigate molecular mechanisms underlying spaceflight-induced brain adaptation and neurodegeneration.

Major Research Projects

Project 1. Spaceflight-induced hippocampal transcriptomic signatures

(ASGSR 2025 Oral Presentation; first-author manuscript under peer review)

- Identified hippocampus-specific transcriptomic signatures associated with oxidative stress, mitochondrial dysfunction, lipid metabolism, neurodegeneration, and ferroptosis using NASA GeneLab RNA-seq datasets following long-duration spaceflight.
- Presented the findings at ASGSR 2025 and developed them into a first-author manuscript currently under peer review.

Project 2. Meta-analysis of regional spaceflight responses and APOE4 susceptibility

(ASGSR 2026 Abstract submitted)

- Integrated pathway-level meta-analysis of multiple NASA GeneLab brain RNA-seq datasets with transcriptomic analyses of human APOE4 and APOE3 iPSC-derived glial transcriptomes to investigate region-specific molecular susceptibility to spaceflight-associated neurodegeneration.

Project 3. Computational workflow development

- Developed OmicsForge, a modular R-based transcriptomics platform for reproducible differential expression analysis, GSEA, transcriptomic meta-analysis, and publication-quality visualization.
- Developed a Python-based preprocessing pipeline for automated Salmon RNA-seq quantification.

2. DOCTORAL RESEARCHER

(Mar. 2019 – Feb. 2025)

Traumatic Brain Injury • Neuroinflammation • Mitochondrial Biology

Investigated molecular and cellular mechanisms underlying secondary brain injury following traumatic brain injury using both in vivo and in vitro experimental models.

Major Research Projects

Project 1. Neuroinflammatory mechanisms following traumatic brain injury

- Investigated the therapeutic effects of heat shock proteins on blood-brain barrier disruption, macrophage infiltration, and M1 macrophage-mediated neuroinflammatory responses following traumatic brain injury.

Project 2. Regulation of astrocyte mitochondrial dysfunction following traumatic brain injury

- Identified two independent HO-1-mediated mechanisms by which Korean Red Ginseng Extract attenuates reactive astrocyte mitochondrial dysfunction following traumatic brain injury.
- The two mechanisms were published as independent first-author research articles.

PUBLICATIONS (SCI ONLY)

1. Til Bahadur Thapa Magar, **Minsu Kim**, Yoonha Kim, Eun Young Choi, Yoon Kyung Choi, Dong-Eun Kim, Kyung Bo Kim, Ji Eun Park, “Pharmacokinetics, Target Engagement of an Immunoproteasome Subunit Low-Molecular-Mass Polypeptide-2 (LMP2) Inhibitor AR-01, and Its Anti-Alzheimer's Effects in Rodents”. *ACS pharmacology & translational science* Jun 29;9(7):1869-1880 (2026).
2. Hyungsu Kim, Sunhong Moon, **Minsu Kim**, Hyungkeun Oh, Jinhong Park, Suji Kim, Taehyung Yoo, Ji-Yoon Kim, Yonghee Kim, Young-Myeong Kim, Yoon Kyung Choi, “Upregulation of astrocytic mitochondrial functions via Korean red ginseng-induced CREB-BK α -HIF-1 α axis through L-type Ca²⁺ channel subunits α 1C and β 4”. *J Cereb Blood Flow & Metabolism*, Feb;46(2):430-443 (2026).
3. Sunhong Moon, Jinseo Park, Sueun Kim, **Minsu Kim**, Hui Su Jeon, Hyungsu Kim, Young-Myeong Kim, ji-yoon Kim, Yoon Kyung Choi, “Korean Red Ginseng-induced astrocytic HIF-1 α : A key regulator of neuroglobin derived from neural stem cell differentiation in physiologic retina and brain”. *Journal of Ginseng Research*, Mar;49(2):189-196 (2025).
4. Eunyong Jung, Ye Eun Kim, Hui Su Jeon, Myeongjong Yoo, **Minsu Kim**, Young-Myeong Kim, Seong-Ho Koh, Yoon Kyung Choi, “Chronic hypoxia of endothelial cells boosts HIF-1 α -NLRP1 circuit in Alzheimer's disease”. *Free Radical Biology and Medicine*, Aug 1;204:385-393 (2023).
5. Hui Su Jeon[†], Chang-Hee Kim[†], **Minsu Kim**, Sunhong Moon, Yoon Kyung Choi, “Korean red ginseng mediates mitochondrial membrane potential repair via the Tom22-Tom20-SIRT2 pathway in astrocytes”. *Conditioning Medicine* Oct, 5(3), 105-111 (2022).
6. Sunhong Moon, Chang-Hee Kim, Jinhong Park, **Minsu Kim**, Hui Su Jeon, Young-Myeong Kim, Yoon Kyung Choi, “Induction of BVR-A Expression by Korean Red Ginseng in Murine Hippocampal Astrocytes: Role of Bilirubin in Mitochondrial Function via the LKB1-SIRT1-ERR α Axis”. *Antioxidants (Basel)*, Sep 1;11(9):1742 (2022).
7. Jinhong Park, Minjae Lee, **Minsu Kim**, Sunhong Moon, Seunghee Kim, Sueun Kim, Seong-Ho Koh, Young-Myeong Kim, and Yoon Kyung Choi, “Prophylactic role of Korean Red Ginseng in astrocytic mitochondrial biogenesis through HIF-1 α ”. *Journal of Ginseng Research*, May;46(3):408-417 (2022).
8. **Minsu Kim**, Sunhong Moon, Hui Su Jeon, Sueun Kim, Seong-Ho Koh, Mi-Sook Chang, Young-Myeong Kim, Yoon Kyung Choi, “Dual Effects of Korean Red Ginseng on Astrocytes and Neural Stem Cells in Traumatic Brain Injury: The HO-1-Tom20 Axis as a Putative Target for Mitochondrial Function”. *Cells* Mar 4;11(5):892 (2022).
9. **Minsu Kim**, Joohwan Kim, Sunhong Moon, Bo Young Choi, Sueun Kim, Hui Su Jeon, Sang Won Suh, Young-Myeong Kim, Yoon Kyung Choi, “Korean Red Ginseng Improves Astrocytic Mitochondrial Function by Upregulating HO-1-Mediated AMPK α -PGC-1 α -ERR α Circuit after Traumatic Brain Injury”. *International Journal of Molecular Sciences*, Dec 3;22(23):13081 (2021).
10. **Minsu Kim**, Hyejung Mok, Woon-Seok Yeo, Joong-Hoon Ahn, and Yoon Kyung Choi, “Role of ginseng in the neurovascular unit of neuroinflammatory diseases focused on the blood-brain barrier”. *Journal of Ginseng Research*, Sep;45(5):599-609 (2021).

MANUSCRIPTS (IN PREPARATION / UNDER REVIEW)

1. **Minsu Kim**, Hyungkeun Oh, Wonjin Park, Tae-Kyeong Lee, Joohwan Kim, Joon Ha Park, Moo-Ho Won, Dong-Eun Kim, Ji-Yoon Kim, Young-Guen Kwon, Ken Arai, Eng H Lo, Young-Myeong Kim, and Yoon Kyung Choi, “Korean red ginseng protects endothelial cells by reducing reactive nitrogen species via HO-1/CO-mediated induction of astrocytic Hsp70 in traumatic brain injury”. Manuscript under review.
2. **Minsu Kim**, Isaac Kremsky, Michael Pecaut, and Xiao Wen Mao, “A Ferroptosis-Related Transcriptional Signature in the Brain After a 35-day Spaceflight”. Manuscript under peer-review.

INTERNATIONAL CONFERENCE

1. **Minsu Kim**, “Astrocyte Ferroptosis in the Brain After Long-Term Spaceflight: Insights from NASA GeneLab RNA-seq Data”. *American Society of Gravitational and Space Research*. Dec 3-6, 2025, Phoenix, AZ, USA. (Oral Presentation)